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PID-based attitude control of quadrotor using robust pole assignment and LPV modeling

Mohammad Hossein Kazemi[1] · **Reza Tarighi**[2]

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Abstract

A novel control strategy is addressed in this paper for tracking control of a Quadrotor attitude based on PID structure. The nonlinear model of quadrotor including gyroscopic effects is considered for a Polytopic Linear Parameter Varying (LPV) modeling. Specifically, the control scheme adopted combines three techniques. On the one hand, the convex LPV controller using the nonlinearity approach has the advantage of precisely describing the system's nonlinear behavior by interpolating local linear models, and one can extrapolate linear controller designs to the nonlinear case considering a more realistic scenario. The principle component analysis (PCA)-based parameter set mapping is used for creating a low dimensional Polytopic LPV model. Augmenting the proposed PID-based control action states to the plant states results in an overall plant in the form of standard robust state feedback control problem where a robust pole assignment controller is designed. An H∞ attenuation level for the tracking error under external disturbances such as wind effects is guaranteed by solving the introduced linear matrix inequalities (LMIs) to compute the control gains. Simulating the planned controller to a proposed quadrotor shows the effectiveness of the proposed strategy in attitude control problem against the external disturbances.

Keywords LMI · Polytopic LPV modeling · Quadrotor dynamics · Robust control

1 Introduction

Although, small Quadrotors, are not being more inefficient from an energetic point of view compared with other aerial vehicles such as conventional helicopter, they have a simpler design, hover capability in a fixed point, perpendicular take-off and landing, and high dexterity and maneuverability.

However, in return for these advantages, they are difficult to control. Because they have fast dynamics and are exposed to severe aerodynamic effects. In addition, the hard restrictions imposed by the propellers, along with various atmospheric disturbances, make more challenging control problem.

✉ Mohammad Hossein Kazemi
kazemi@shahed.ac.ir

Reza Tarighi
Tarighi_r@proton.mail.com; s_r_tarighi@azad.ac.ir

1 Electrical Engineering Department, Shahed University, Persian Gulf Hwy, Tehran 33191-18651, Iran

2 Department of Control Engineering, Faculty of Electrical Engineering, South Tehran Branch, Islamic Azad University, Tehran, Iran

Many samples of quadratic applications can be seen in the real world such as high risk places disasters, and rescue [1–3] agriculture [4, 5], facilities inspection [6], traffic surveillance [7] or photogrammetry and mapping [8]. So far, a lot of work has been done in the field of quadrotor control. They have introduced the control design and development with different types of controllers, such as multi-agent system and formation control [9–12], linear quadratic regulator [13–15], feedback linearization, model predictive, sliding mode control [16, 17], backstepping control approach, [18–20], (LPV) strategies [21], as well as learning approaches like [22], or MPC-based deep learning [23].

In [9], cooperative formation control of Quadrotors with obstacle avoidance and self-collisions based on a hierarchical MPC approach have been presented. In [10], adaptive leader follower control for multiple Quadrotors via multiple surfaces control is provided. Based on the different surface and settled undirected communication topology, the versatile pioneer adherent control for numerous quadrotors is talked about. Cooperative control of numerous UAVs for timberland fire observing and location is presented in [11]. In [12], a study on the arrangement control of numerous Quadrotors has been introduced. This reference points out the challenges

and the propensity of investigate in the field of different quadrotors control.

Quadrotor is a six-degree-of-freedom (DOF) systems with three degrees of transitional freedom of movement and three degrees of angular freedom, while having only four control actions consisting of an entire lifting force and three torques created by variating the speeds of the motors. Generally, the control of a quadrotor can be divided in two control problems: position control as an outer loop and attitude control as an inner loop with faster dynamics. The inner attitude control loop tracks the commanded roll and pitch angles coming from the outer position control loop, plus an externally enforced yaw angle, by the above mentioned three torques[24–28], In [24], a robust cascade controller counting a demeanor controller and a position controller is proposed based on the progressive control conspire and the vigorous compensating method. The model show is depicted as a MIMO, a time-varying framework subject to parametric irritations, nonlinear and coupled elements, outside unsettling influences, and state and input delays. In [25], the utilization of a LPV representation of the demeanor subsystem permits to consider it as free of position one, permitting the utilization of a cascade control structure, and also for the internal demeanor control circle, the proposed qLPV–LQR controller, and proportional integral observer are unraveled in an ideal way by fathoming a set of LMIs. A super turning sliding mode controller is outlined for a Quadrotor by using a cascaded internal external circle structure in [26]. The controller is tried against wind turbulence conditions and displaying vulnerabilities. It is compared with a commonly used LQR-PD controller and a nonlinear criticism linearization-based controller. In [27], the robust adaptive control scheme based on back-stepping technique is presented that improves the trajectory tracking performance of the Quadrotor. A new NLPID controller for stabilizing and controlling a 6-DOF UAV system is proposed in [28].

Therefore, the attitude control is of great importance that has attracted more focus in the literature, Since the attitude maneuvers are highly sensitive to external disturbances [29, 30], considering the problem of external disturbances attenuation during attitude control design is an essential purpose [31]. In [32], the problem of attitude control is examined by presenting a fuzzy adaptive sigmoid generalized super-twisting extended state observer to estimate and to compensate the disturbance with different upper bounds of the first derivative. The sliding mode controller can also be used as an alternative method for disturbances rejection [26]. Some researchers have proposed the disturbance observer-based control techniques for quadrotor attitude toward external disturbances [30].

Polytopic linear parameter varying (LPV) modeling with a not stationary behavior [33] can be a convenient operative way to deal with nonlinear systems, even systems whose nonlinearities are state-dependent, such as quadrotors. A review on different outlines of the LPV control synthesis practices is accessible in [34]. In [35], polytopic attitude control using a nonlinear model of an unmanned helicopter is introduced and presented. The basis of linearization scheduling, formed by applying Jacobian of the system equations around the operating points, can be used to obtain the polytopic model of a nonlinear system [36]. Quadratic Lyapunov functions that are affine in the parameter proposed in [37] to verify robust stability, robust convergence rates, and robust performance of the LPV systems in terms of LMI feasibility tests.

In [38], a state-feedback proportional–integral control is offered for LPV systems which is implemented to control of longitudinal and lateral dynamics of an aircraft and also in [39], trajectory tracking with state-feedback for rotorcraft with LPV is presented. Affine LPV systems under Polytopic model uncertainty are considered in [40], where the structure of the controller is centralized/decentralized fixed order output feedback. The proposed controller is a gain-scheduled PID designed in the form of bilinear matrix inequality (BMI). In [41], a scheduled PID is designed with a hybrid synthesis including LMI techniques and evolutionary search approaches applied to an affine LPV model of a spark-ignition engine. In [42] a gain-scheduled PID controller with Smith predictor is introduced to control the open-flow channel by means of LPV modeling. It is expressed as a convex state-feedback problem that involves an LMI-based pole placement problem. The proposed $H\infty$ control in [43] deals with longitudinal and lateral directions. Although its capability cannot be denied, there is no enlightenment on how external disturbances affect as well as simultaneous interaction in the longitudinal and transverse directions. The attitude control is studied in [44] using robust LPV for a Quadcopter considering state constraints and input saturation. The scheduling parameters, which consists of some states and the control inputs, must vary in a specified polyhedron. However, since the speeds of propellers are considered as a number of scheduling signal, a heavy computational load and complex calculation are some of the most important restrictions of the usage of the proposed method. Introduced quasi-LPV modeling in [45], is examined based on changes in Quadrotor mass, rotor speed and the moments of inertia. The proposed attitude controller consists of two LPV-$H\infty$ controllers. The first is to control the roll and pitch angles, and the second is to control the yaw angle. In [46], modeling and control design is investigated using output feedback and considering the motor dynamics. It introduces a simple LPV modeling that includes only two varying parameters.

In [47], modeling and identification of the nonlinear twin rotor system have been performed applying quasi-LPV concepts. The system is approximated to a Polytopic model for designing a pole placement controller in the form of

state feedback. The main drawback is the high volume of computations due to the large number of parameters that increase the LMIs, although the paper uses only the maximum and minimum values of the parameters variations are used to determine the Polytopic vertices. ANA robust Polytopic state-feedback is proposed in [24] for the control of quadrotor, which counteracts the adverse effects of mass deviations and battery voltage variants. It has only four vertices and does not include other system parameter variations.

In this article, the issue of quadrotor attitude control has been investigated based on a novel technique using reduced Polytopic LPV modeling and a hybrid PID control action in the form of state-feedback. The control objective includes both $H\infty$ disturbance attenuation and LMI region pole assignment. Although, the proposed controller has a PID structure, its structure is not decentralized form. This means that for each quadrotor Euler angle, there is no separate PID controller, but the controllers operate interactively. In fact, an overall centralized PID controller is designed for whole of the quadrotor angles. Its states are augmented to the plant states and constructing a general linear fractional transformation (LFT) representation. In this work a quasi-LPV modeling is introduced to design a PID-based state-feedback control revealing its main advantage, which is the capability of overcoming the nonlinearity effects of the system. The gyroscopic term is embedded as a varying parameter alongside other variable parameters such as Euler angles, angular velocities, mass and moments of inertia. A set of gridding points is created by considering the minimum, maximum, and middle values of each variable parameter to generate sample linear plants for building an initial Polytopic model. In the next step, due to the high number of vertices in the initial built Polytopic model, a parameter set mapping with the basis of PCA algorithm [48], is used to obtain a reduced Polytopic model by neglecting the less significant directions in the parameter space. When the parametric space dimension is large and the computational complexity may be occurred, this strategy can be actually valuable [49]. The obtained reduced Polytopic model, with the lowest number of vertices, is used for designing the proposed controller. The control objective is to assign the poles of the closed loop system at each vertex inside a prescribed LMI region such that an $H\infty$ attenuation level is guaranteed for the closed loop system under input disturbances. The control gains are specified by solving the related LMIs constraints. The proposed control simulation is performed on nonlinear model of a quadrotor that is expected to confirm the effectiveness of the proposed method.

The main contribution of this paper is summarized to address a control design that combines three techniques to obtain a convex LPV controller for a quadrotor considering the gyroscopic effect: introducing sufficient conditions in the form of LMIs, whose solutions guarantee the control objectives; the PCA-based parameter set mapping is used for creating a low dimensional Polytopic LPV model; an $H\infty$ attenuation level of external disturbances beside a robust pole placement are considered in a PID-based controller.

The paper sections are prepared as follows. Following sections, some preliminaries on LPV systems, introducing of the Quadrotor dynamics, Quasi-LPV representation and Polytopic modeling with PCA algorithm, control design procedure, numerical simulation and comparison with nonlinear PID (NLPID) are discussed. Finally, the article ends with a conclusion in the last section.

2 Preliminaries

Consider a standard closed loop control structure with following state space equations.

$$\dot{x} = A(\mu)x + B_1(\mu)w + B_2(\mu)u \tag{1}$$

$$z_\infty = C(\mu)x + D_1(\mu)w + D_2(\mu)u \tag{2}$$

where $x \in R^n$ is the state vector, $w \in R^{n_w}$ is the exogenous input, $u \in R^{n_u}$ is the control input, and $z_\infty \in R^{n_z}$ is the control objective. The varying parameter $\mu \in R^{n_s}$ is assumed to be in a compact set. It is assumed that all the matrices have appropriate dimensions and they are constrained to the following polytope.

$$\begin{aligned} P = \{&(A,\, B_1,\, B_2,\, C,\, D_1,\, D_2)(\mu) \mid \\ &(A,\, B_1,\, B_2,\, C,\, D_1,\, D_2)(\mu) \\ &= \sum_{i=1}^{N} \alpha_i (A_i,\, B_{1i},\, B_{2i},\, C_i,\, D_{1i},\, D_{2i}), \\ &\sum_{i=1}^{N} \alpha_i = 1, \quad \alpha_i \geq 0,\ i = 1, 2, ..., N\} \end{aligned} \tag{3}$$

In other words, the system matrices depend on the time-varying vector $\mu(t) = [\,\mu_1(t) \ldots \mu_N(t)\,]^T$.The vertices of the polytope P, referred as $(A_i,\, B_{1i},\, B_{2i},\, C_i,\, D_{1i},\, D_{2i})$ for $i = 1, 2, ..., N$. The H_∞ performance achievement is defined to synthesis a control signal for (1) that provides the smallest attenuation level $\gamma > 0$ such that for any external input $w \in L_2$, it is verified that $z_\infty \in L_2$, and:

$$\|z_\infty\|_2 < \gamma \|w\|_2 \tag{4}$$

The following lemma gives the sufficient condition of existence a state-feedback gain matrix K for stabilizing the closed loop system assuring the attenuation level γ.

Lemma 1 [50–52] If there exist a symmetric positive definite matrix X and a matrix Y with appropriate dimensions such that the following LMIs are satisfied,

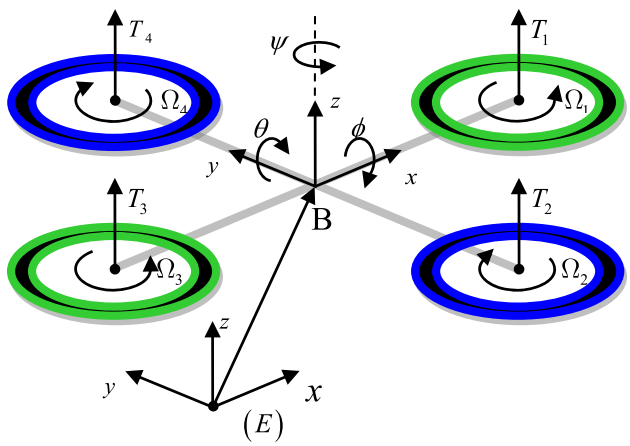

Fig. 1 Quadrotor and its reference frames

$$\begin{bmatrix} A_i X + X A_i^T + B_{2i} Y + Y^T B_{2i}^T & * & * \\ B_{1i}^T & -I & * \\ C_i X + D_{2i} Y & D_{1i} & -\gamma^2 I \end{bmatrix} < 0, \quad (5)$$

$i = 1, 2, \ldots, N$

where symbol $*$ indicates symmetric blocks in the LMIs, then the state feedback control gain $K = YX^{-1}$ quadratically stabilizes the uncertain system (1), (2) with disturbance attenuation γ.

3 Quadrotor dynamics

Mathematical modeling of quadrotor UAV using Newton Euler approach has been explained in many researches [25, 53]. Quadrotors as an air vehicle have four propellers, two rotating clockwise and the other two rotating counterclockwise. The movement of the quadrotor is due to the rotational motion of the propellers, such that by increasing or decreasing four propellers speeds all together, a vertical motion occurs while differing the speed of opposite propellers causes a roll or pitch motion. The yaw motion is also caused by torque imbalance between each pair of propellers. Suppose that for the quadrotor shown in Fig. 1, $[\phi, \theta, \psi]^T$, $[x, y, z]^T$, $[u, v, w]^T$, and $[p, q, r]^T$ denote the Euler angles, the position in earth frame, linear velocity, and angular velocity in body frame, respectively. Assuming the rigidity and symmetricity of the body as well as nearing to hover attitude, the quadrotor dynamic model in earth frame and with considering wind torque vector $\tau_w = [\tau_{wx}, \tau_{wy}, \tau_{wz}]^T$ is described by the following equations [54, 55].

Table 1 Quadrotor parameters

Symbol	Description	Value
I_x	Moment of inertia along x-axis	3.4×10^{-3} N m s^2
I_y	Moment of inertia along y-axis	3.4×10^{-3} N m s^2
I_z	Moment of inertia along z-axis	6×10^{-3} N m s^2
m	The quadrotor mass	0.698 kg
g	Gravitational Acceleration	9.81 m s^{-2}
J_{TP}	Total rotational moment of inertia along propeller axis	1.3×10^{-6} N m s^2
ℓ	Central distance between the quadrotor gravity and each motor	0.171 m
c_T	Thrust factor	7.62×10^{-8} $\frac{\text{N}}{\text{rpm}^2}$
c_Q	Drag factor	2.68×10^{-9} $\frac{\text{N}}{\text{rpm}^2}$
$\boldsymbol{\Omega}_{\min}$	Minimum propellers speeds	1075 rpm
$\boldsymbol{\Omega}_{\max}$	Maximum propellers speeds	8600 rpm

$$\begin{aligned}
\ddot{x} &= (\cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi)\frac{U_1}{m} \\
\ddot{y} &= (\cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi)\frac{U_1}{m} \\
\ddot{z} &= -g + (\cos\phi \cos\theta)\frac{U_1}{m} \\
\ddot{\phi} &= \dot{\theta}\dot{\psi}\frac{I_y - I_z}{I_x} - \frac{J_{TP}}{I_x}\dot{\theta}\,\boldsymbol{\Omega}_r + \frac{U_2 + \tau_{wx}}{I_x} \\
\ddot{\theta} &= \dot{\phi}\dot{\psi}\frac{I_z - I_x}{I_y} + \frac{J_{TP}}{I_y}\dot{\phi}\,\boldsymbol{\Omega}_r + \frac{U_3 + \tau_{wy}}{I_y} \\
\ddot{\psi} &= \dot{\phi}\dot{\theta}\frac{I_x - I_y}{I_z} + \frac{U_4 + \tau_{wz}}{I_z}
\end{aligned} \quad (6)$$

The control inputs, trust force U_1 and three roll, pitch and yaw torques (U_2, U_3, U_4) are related to the rotors speed ($\boldsymbol{\Omega}_i$, $i = 1, 2, 3, 4$), as follows:

$$\begin{aligned}
U_1 &= c_T(\boldsymbol{\Omega}_1^2 + \boldsymbol{\Omega}_2^2 + \boldsymbol{\Omega}_3^2 + \boldsymbol{\Omega}_4^2) \\
U_2 &= c_T.\ell(-\boldsymbol{\Omega}_2^2 + \boldsymbol{\Omega}_4^2) \\
U_3 &= c_T.\ell(-\boldsymbol{\Omega}_1^2 + \boldsymbol{\Omega}_3^2) \\
U_4 &= c_Q(-\boldsymbol{\Omega}_1^2 + \boldsymbol{\Omega}_2^2 - \boldsymbol{\Omega}_3^2 + \boldsymbol{\Omega}_4^2)
\end{aligned} \quad (7)$$

All the parameters in the equations as well as the values considered for them are presented in Table 1, as well as, the cascaded controller architecture is presented in Fig. 2. The parameter values for performing the simulations are taken

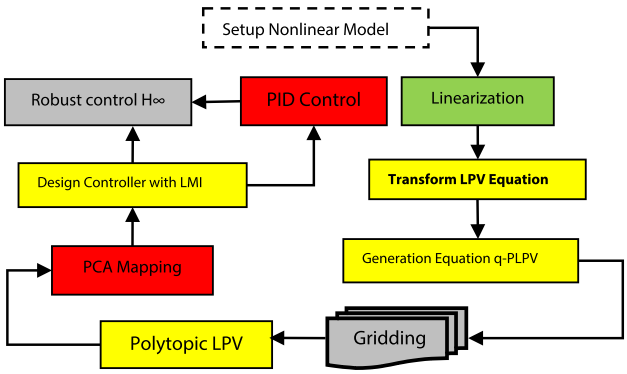

Fig. 2 Cascaded controller architecture

from [25]. The term $\boldsymbol{\Omega}_r$ which appears in the gyroscopic effect, is due to an imbalance in the total algebraic sum of the propellers speeds.

$$\boldsymbol{\Omega}_r = -\boldsymbol{\Omega}_1 + \boldsymbol{\Omega}_2 - \boldsymbol{\Omega}_3 + \boldsymbol{\Omega}_4 \tag{8}$$

In the control of robotic systems such as Quadrotor, which have complex dynamics, we usually ignore the dynamics of the motors against the dynamics of the robot. The reason is that we work in a higher level control layer whose purpose is to specify the torque required to satisfy the control objectives. The lower control layer is responsible for providing this torque. If it is assumed that this control layer performs its task correctly, i.e., the motors are equipped with high-speed torque control hardware compared to the dynamics of the robot; and then, the dynamics of the motors can be omitted.

3.1 Quasi-LPV representation and Polytopic modeling

In this work, the attitude control of quadrotor has been studied by adjusting the Euler angles according to the desired reference inputs. The attitude subsystem includes the last three equations of (6), which means the dynamical equations of Euler angles with three inputs U_2, U_3, and U_4. Since the parameter varying are state-depended, we are faced with a quasi-LPV modeling. This allows us to embed the quadrotor nonlinearities into the varying parameters and therefore not only the nonlinearities are disappeared, but also get rid of the gyroscopic effect. The scheduling signals ρ_i are related to the varying parameters of the system according to Table 2. Their variation ranges are also presented in this table.

Thus, defining $x_{a1} := [\phi, \theta, \psi]^T$ and $x_{a2} := [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$ the state space representation of the attitude subsystem can be expressed in a quasi-LPV form as follows:

Table 2 Scheduling signals

ρ	Symbol	Variation Range
ρ_1	I_x	$\pm 10\%$
ρ_2	I_y	$\pm 10\%$
ρ_3	I_z	$\pm 10\%$
ρ_4	J_{TP}	$\pm 10\%$
ρ_5	$\dot{\phi}$	$[-2\ +2]$ rad s^{-1}
ρ_6	$\dot{\theta}$	$[-2\ +2]$ rad s^{-1}
ρ_7	$\boldsymbol{\Omega}$	$[-100\ +100]$ rad s^{-1}

$$\begin{aligned} \dot{x}_{a1} &= x_{a2} \\ \dot{x}_{a2} &= \begin{bmatrix} 0 & \mu_1 & \mu_2 \\ \mu_3 & 0 & \mu_4 \\ \mu_5 & \mu_6 & 0 \end{bmatrix} x_{a1} + \begin{bmatrix} \mu_7 & 0 & 0 \\ 0 & \mu_8 & 0 \\ 0 & 0 & \mu_9 \end{bmatrix} \begin{bmatrix} U_2 + \tau_{wx} \\ U_3 + \tau_{wy} \\ U_4 + \tau_{wz} \end{bmatrix} \end{aligned} \tag{9}$$

whereμ_i's are the varying parameters with the respect of scheduling signals as follows:

$$\begin{aligned} \mu_1 &= -\tfrac{\rho_4}{\rho_1}\rho_6 & \mu_6 &= \tfrac{\rho_1-\rho_2}{2\rho_3}\rho_5 \\ \mu_2 &= \tfrac{\rho_2-\rho_3}{\rho_1}\rho_6 & \mu_7 &= \tfrac{1}{\rho_1} \\ \mu_3 &= \tfrac{\rho_4}{\rho_2}\rho_7 & \mu_8 &= \tfrac{1}{\rho_2} \\ \mu_4 &= \tfrac{\rho_3-\rho_1}{\rho_2}\rho_5 & \mu_9 &= \tfrac{1}{\rho_3} \\ \mu_5 &= \tfrac{\rho_1-\rho_2}{2\rho_3}\rho_6 \end{aligned} \tag{10}$$

By considering the $x_a := [x_{a1}^T, x_{a2}^T]^T$ as the attitude state vector, the quasi-LPV can be summarized to:

$$\dot{x}_a = A_a(\mu)x_a + B_a(\mu)\tau_w + B_a(\mu)u_a \tag{11}$$

where $u_a = [U_2, U_3, U_4]^T$ is the input vector, $\mu = [\mu_1, \ldots, \mu_9]^T$ is the varying parameter vector, and the matrices $A_a(\mu)$ and $B_a(\mu)$ are as follows:

$$A_a(\mu) = \begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & A_{a22}(\mu) \end{bmatrix}, \quad B_a(\mu) = \begin{bmatrix} 0_{3\times3} \\ B_{a2}(\mu) \end{bmatrix} \tag{12}$$

and,

$$A_{a22}(\mu) = \begin{bmatrix} 0 & \mu_1 & \mu_2 \\ \mu_3 & 0 & \mu_4 \\ \mu_5 & \mu_6 & 0 \end{bmatrix}, \quad B_{a2}(\mu) = \begin{bmatrix} \mu_7 & 0 & 0 \\ 0 & \mu_8 & 0 \\ 0 & 0 & \mu_9 \end{bmatrix} \tag{13}$$

In order to construct a Polytopic model, we consider only three values for each scheduling signal: minimum, middle,

and maximum of its interval range. In this way, $N = 3^{\overline{\rho}}$ vertices are produced, where $\overline{\rho} = 7$ is the number of scheduling signals, and the vertices are noted by:

$$(A_{ai},\ B_{ai}) := \left(\begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & A_{a22i} \end{bmatrix}, \begin{bmatrix} 0_{3\times3} \\ B_{a2i} \end{bmatrix} \right) \tag{14}$$

while we have 9 parameters, and $2^{\overline{\mu}}$ vertices, where $\overline{\mu} = 9$ is the number of varying parameters, are needed to make a Polytopic model for the system. In any case, the high number of vertices requires a high computational volume, and therefore, in the next section we will reduce the number of vertices using the PCA-based parameter set mapping technique. However, the attitude subsystem will be constrained to the following initial polytope.

$$\begin{aligned} P_a &= \{(A_a,\ B_a)(\mu) \mid (A_a,\ B_a)(\mu) \\ &= \sum_{i=1}^{N} \alpha_i (A_{ai},\ B_{ai}),\ \ \sum_{i=1}^{N} \alpha_i = 1\} \end{aligned} \tag{15}$$

The vertices of the polytope P_a, referred as $(A_{ai},\ B_{ai})$ for $i = 1, 2, ..., N$. The matrices, are continuous functions, in terms of parameter vector $\mu(t) \in \Re^{\overline{\mu}}$, depends on the scheduling signal $\rho(t) \in \Re^{\overline{\rho}}$ according to $\mu(t) = h(\rho(t))$ where $h : \Re^{\overline{\rho}} \to \Re^{\overline{\mu}}$ is a continuous parameter mapping as (10).

3.2 PCA Algorithm for LPV Models

In this section, parameter set mapping based on PCA algorithm is used to find tighter regions in the space of the scheduling parameters. By neglecting insignificant directions in the mapped parameter space, approximations of LPV models are achieved which will lead to a less conservative controller synthesis.

The number of gridding points, N, is chosen sufficiently large concern to system operating range, nonlinearity effects, and system parameters dimension such that all dynamic behavior of the quadrotor is captured. On the other hand, increasing N, causes high complexity in controller computations. To find a tighter region in the parameter space, parameter set mapping on the basis of PCA algorithm is implemented. The Polytopic LPV model (15) is simplified to a space with lower dimension by ignoring the slight directions in the mapped parameter space. The procedure is explained in the following accordance with the procedure described in [48]. For the LPV model (11) and scheduling signal $\rho(t)$, the mapping $\eta(t) = r(\rho(t)),\ \ r : \Re^{\overline{\rho}} \to \Re^{\overline{\eta}}$ where $\overline{\eta} < \overline{\mu}$, should be found such that

$$\dot{x}_a = \hat{A}_a(\eta)x_a + \hat{B}_a(\eta)u_a + \hat{B}_a(\eta)\tau_w \tag{16}$$

Approximates the model (11). The basic details of PCA can be studied in [56]. The parameters of the Polytopic LPV model (15) for $i = 1,\ 2,\ \ldots,\ N$ are gathered to construct a $\overline{\mu} \times N$ data matrix $\Theta = [\mu^1,\ \mu^2,\ \ldots,\ \mu^N]$, where μ^i is the varying parameter vector evaluated at i–th vertex. Normalizing its rows with zero mean and unit standard deviation to constructing the normalized data matrix $\Theta^n = \Pi(\Theta)$. Next, performing the following singular value decomposition

$$\Theta^n = \begin{bmatrix} \hat{U} & U \end{bmatrix} \begin{bmatrix} \hat{\Sigma} & 0 \\ 0 & \Sigma \end{bmatrix} \begin{bmatrix} \hat{V}^T \\ V^T \end{bmatrix} \tag{17}$$

and, separating the $\overline{\eta}$ significant singular values corresponding to $\hat{U}$, $\hat{\Sigma}$, and $\hat{V}$. Ignoring the lower singular values leads to $\hat{\Theta}^n = \hat{U}\hat{\Sigma}\hat{V}^T \approx \Theta^n$ where $\hat{\Theta}^n$ is approximation of the normalized given data Θ^n. The matrix $\hat{U}$ is used as a basis of the significant column space to realize the reduced mapping

$$\eta(t) = r(\rho(t)) = \hat{U}^T \Pi(h(\rho(t))) = \hat{U}^T \Pi(\mu(t)) \tag{18}$$

It means, the approximate mapping of $\hat{A}_a(\cdot)$ and $\hat{B}_a(\cdot)$ in (16) are related to (11) by:

$$\begin{aligned} \hat{P}_a(\eta) &= \begin{bmatrix} \hat{A}_a(\eta(t)) & \hat{B}_a(\eta(t)) \end{bmatrix} \\ &= \begin{bmatrix} A_a(\hat{\mu}(t)) & B_a(\hat{\mu}(t)) \end{bmatrix} \end{aligned} \tag{19}$$

where

$$\hat{\mu}(t) = \Pi^{-1}(\hat{U}\phi(t)) = \Pi^{-1}(\hat{U}\hat{U}^T \Pi(\mu(t))) \tag{20}$$

Note that Π^{-1} indicates the row-wise rescaling. Thus, using $\hat{\Theta}^n$ to regenerate the new vertices $(\hat{A}_{ai},\ \hat{B}_{ai})$, the Polytopic LPV model (15) is reduced to:

$$\begin{aligned} \hat{P}_a &= \left\{(A_a,\ B_a)(\hat{\mu}) \mid (A_a,\ B_a)(\hat{\mu}) \right. \\ &= \sum_{i=1}^{\hat{N}} \alpha_i (\hat{A}_{ai},\ \hat{B}_{ai}) \\ &\left. ,\ \sum_{i=1}^{\hat{N}} \alpha_i = 1,\ \ \alpha_i \geq 0,\ \ i = 1, 2, ..., \hat{N} \right\} \end{aligned} \tag{21}$$

where $\hat{N} = 2^{\overline{\eta}}$ is the number of vertices and $\hat{P}_{ai} = P_a(\hat{\mu}_{vi}) := (\hat{A}_{ai},\ \hat{B}_{ai})$ is the i-th new vertex model such that:

$$(\hat{A}_{ai},\ \hat{B}_{ai}) := \left(\begin{bmatrix} 0_{3\times3} & I_3 \\ 0_{3\times3} & \hat{A}_{a22i} \end{bmatrix}, \begin{bmatrix} 0_{3\times3} \\ \hat{B}_{a2i} \end{bmatrix} \right) \tag{22}$$

The following fraction of the total variation $v_{\overline{\eta}}$[48], can be introduced as a criterion for measuring the quality of the

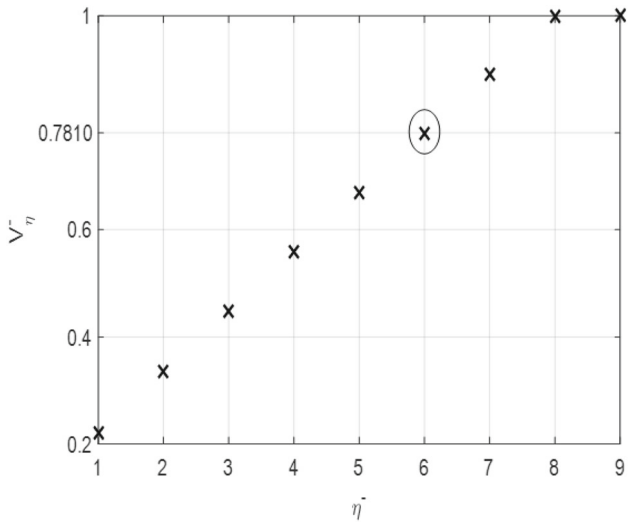

Fig. 3 Fraction of total variation $v_{\overline{\eta}}$ versus $\overline{\eta}$

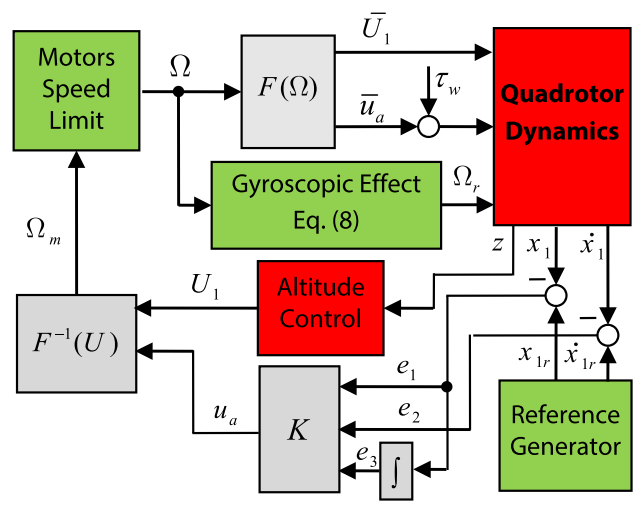

Fig. 4 Proposed control structure

approximated Polytopic model (21) relative to actual Polytopic model (15) by the singular values in (17).

$$v_{\overline{\eta}} = \frac{\sum_{i=1}^{\overline{\eta}} \sigma_i^2}{\sum_{i=1}^{\overline{\mu}} \sigma_i^2} \tag{23}$$

It should be noted that the fraction of the total variation is used as a criterion for measuring the approximation quality, obviously a lower approximation may compromise the feasibility.

For selecting the appropriate value of $\overline{\eta}$, the relation (23) is depicted in Fig. 3 for different values of $\overline{\eta}$. Then the suitable $\overline{\eta}$ is chosen in terms of desired acceptable percent of the error in captured information. Here, choosing $\overline{\eta} = 6$, about 78% of the information is captured and the reduced Polytopic LPV model (21) will have only $\hat{N} = 2^6 = 64$ vertices.

3.3 Control strategy

The main purpose of the attitude control system is for the Euler angles x_{a1} to track the input reference $x_{1r} := [\phi_r, \theta_r, \psi_r]^T$, so that the tracking error $e_1 := x_{1r} - x_{a1}$, for the reduced Polytopic LPV model (21), converges to zero, asymptotically. Our proposed strategy is adapted from the structure of a typical PID controller, where

$$u_a = K_P e_1 + K_D \dot{e}_1 + K_I \int_0^t e_1(\sigma) d\sigma \tag{24}$$

Defining error vector $e := [e_1^T, e_2^T, e_3^T]^T$ with $e_2 := \dot{e}_1$ and $e_3 := \int_0^t e_1(\sigma)\, d\sigma$, and also $z_\infty := \alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3$ where α_i's are designing parameters, as an objective signal for H_∞ performance improvement, the state equation of the error system are as follows:

$$\begin{aligned}
\dot{e}_1 &= e_2 = \dot{x}_{1r} - \dot{x}_{a1} = \dot{x}_{1r} - x_{a2} \\
\dot{e}_2 &= \ddot{e}_1 = \ddot{x}_{1r} - \dot{x}_{a2} - \\
&= \ddot{x}_{1r} - A_{a22}(\hat{\mu}) x_{a2} - B_{a2}(\hat{\mu})(u_a + \tau_w) \\
&= A_{a22}(\hat{\mu}) e_2 - B_{a2}(\hat{\mu})(u_a + \tau_w) \\
&\quad + \ddot{x}_{1r} - A_{a22}(\hat{\mu}) \dot{x}_{1r} \\
\dot{e}_3 &= e_1
\end{aligned} \tag{25}$$

The above error sate equations can be rewrite in the following compact form:

$$\begin{aligned}
\dot{e} &= A_e(\hat{\mu}) e + B_{e1}(\hat{\mu}) d_w + B_{e2}(\hat{\mu}) u_a \\
z_\infty &= C_a e
\end{aligned} \tag{26}$$

where d_w as follows is the total perturbation vector due to wind torque and differential deviations of the reference input, and:

$$\begin{aligned}
&A_e(\hat{\mu}) = \begin{bmatrix} 0 & I & 0 \\ 0 & A_{a22}(\hat{\mu}) & 0 \\ I & 0 & 0 \end{bmatrix}, \quad B_{e1}(\hat{\mu}) = \begin{bmatrix} 0 & 0 \\ I & -B_{a2}(\hat{\mu}) \\ 0 & 0 \end{bmatrix}, \\
&B_{e2}(\hat{\mu}) = \begin{bmatrix} 0 & -B_{a2}^T(\hat{\mu}) & 0 \end{bmatrix}^T, \\
&C_a = \begin{bmatrix} \alpha_1 I & \alpha_2 I & \alpha_3 I \end{bmatrix}, \; d_w = \begin{bmatrix} \ddot{x}_{1r} - A_{a22}(\hat{\mu}) \dot{x}_{1r} \\ \tau_w \end{bmatrix}
\end{aligned} \tag{27}$$

Now, it can be seen that control law (24) is equivalent to a full state-feedback control law $u_p = Ke$ for (26), where $K = [K_P, K_D, K_I]$. The control structure is shown in Fig. 4. The

Euler angles as Quadrotor's attitude and their derivatives are measured and compared with references; and then, the error signals and the integral of the attitude error are multiplied by the control gain K to obtain control actions.

The control action signals should first be converted to the motors speed commands by inverse of (7) (shown with $F^{-1}(U)$) and after applying the motors speed limits and ignoring the motors dynamics, the actual angular torques and the gyroscopic effect are computed by (7) (shown with $F(\boldsymbol{\Omega})$) and (8) respectively, and then applied to the Quadrotor dynamics. Note that since the trust force U_1 is effective in speed calculations, we have to use a conventional altitude controller to keep the Quadrotor at an almost constant height. We used a simple PID controller for this purpose.

The state-feedback gain matrix K for stabilizing the closed loop system assuring the H_∞ attenuation level γ is computed by solving the LMIs that are introduced in the following theorem.

Theorem 1 If there exist a symmetric positive definite matrix X and a matrix Y with appropriate dimensions such that the following LMIs are satisfied,

$$\begin{bmatrix} Q_{ai} & * & * \\ \hat{B}_{a1i}^T & -I & * \\ C_a X & D_1 & -\gamma^2 I \end{bmatrix} < 0, \quad i = 1, 2, ..., \hat{N} \tag{28}$$

where

$$\begin{aligned} Q_i := & \begin{bmatrix} 0 & I & 0 \\ 0 & \hat{A}_{a22i} & 0 \\ I & 0 & 0 \end{bmatrix} X + X \begin{bmatrix} 0 & I & 0 \\ 0 & \hat{A}_{a22i} & 0 \\ I & 0 & 0 \end{bmatrix}^T \\ & + \begin{bmatrix} 0 \\ -\hat{B}_{a2i} \\ 0 \end{bmatrix} Y + Y^T \begin{bmatrix} 0 \\ -\hat{B}_{a2i} \\ 0 \end{bmatrix}^T \end{aligned} \tag{29}$$

and,

$$\hat{B}_{a1i} := \begin{bmatrix} 0 & 0 \\ I & -\hat{B}_{a2i} \\ 0 & 0 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 0 & 0 \end{bmatrix} \tag{30}$$

Suppose that all matrices have the appropriate dimensions. Then, the control law (24) with control gain $K = YX^{-1}$ quadratically stabilizes the uncertain system error system (26) with disturbance attenuation γ.

Proof the proof is straightforward by applying the Lemma 1 to the uncertain error system (26). If the following substitution is made in Eqs. (1) and (2):

$$\begin{aligned} & A(\mu) = A_e(\hat{\mu}), \quad B_1(\mu) = B_{e1}(\hat{\mu}) \\ & B_2(\mu) = B_{e2}(\hat{\mu}), \quad C(\mu) = C_a, \end{aligned}$$

Then the LMI (5) is converted to the LMI (28) and the theorem is proved. □

To improve the performance of the controller, a pole placement objective is added to the previous controller. Therefore, the LMI region $\mathcal{D}$, a 'conic sector' of the complex plane with inner angle about 139°, it means and apex at the $(-2\beta, 0)$, is considered for assigning the poles of the closed loop Polytopic LPV model. The LMI region $\mathcal{D}$ is described by:

$$\begin{aligned} & \mathcal{D} = \{z \in \mathcal{C} : \quad f_{\mathcal{D}}(z) < 0\} \\ & f_{\mathcal{D}}(z) = L + zM + \bar{z}M^T \end{aligned} \tag{31}$$

where matrices L and M are:

$$L = \begin{bmatrix} \beta & 0 \\ 0 & \beta \end{bmatrix}, \quad M = \begin{bmatrix} \sin(\delta) & \cos(\delta) \\ -\cos(\delta) & \sin(\delta) \end{bmatrix} \tag{32}$$

If there exist a symmetric positive definite matrix X and a matrix Y such that the following LMIs are satisfied

$$\begin{bmatrix} \beta X + \sin\delta(V_i + V_i^T) & \cos\delta(V_i - V_i^T) \\ \cos\delta(V_i^T - V_i) & \beta X + \sin\delta(V_i + V_i^T) \end{bmatrix} < 0 \tag{33}$$

where

$$V_i := \begin{bmatrix} 0 & I & 0 \\ 0 & \hat{A}_{a22i} & 0 \\ I & 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 \\ -\hat{B}_{a2i} \\ 0 \end{bmatrix} Y, \tag{34}$$

for $i = 1, ..., \hat{N}$,

Then, the control law (24) with control gain $K = YX^{-1}$ quadratically $\mathcal{D}$-stabilizes the uncertain error system (26) for the desired LMI region (31) [57].

As a result, computing the LMI conditions (28) and (33) for the same variables X and Y result in the control gain $K = YX^{-1}$ with the quadratic H_∞ performance with disturbance attenuation γ that assign the closed loop poles in $\mathcal{D}$.

4 Numerical simulation

This section describes the performance of the proposed control scheme applied to the quadrotor dynamics (6) to examine against the nonlinearity effects, gyroscopic effect, exogenous disturbances and parameter perturbations.

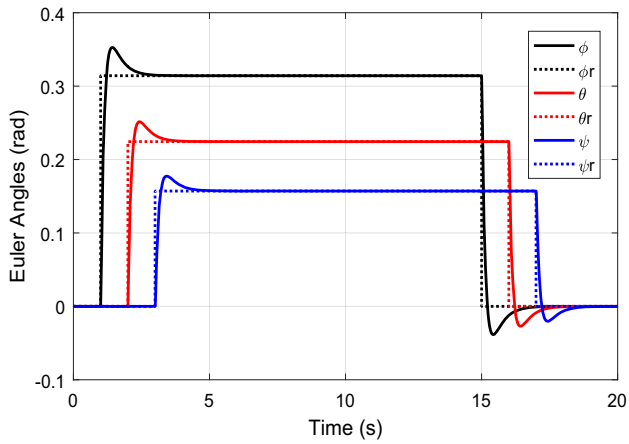

Fig. 5 Attitude system outputs by H_∞ control with attenuation level $\gamma = 1$

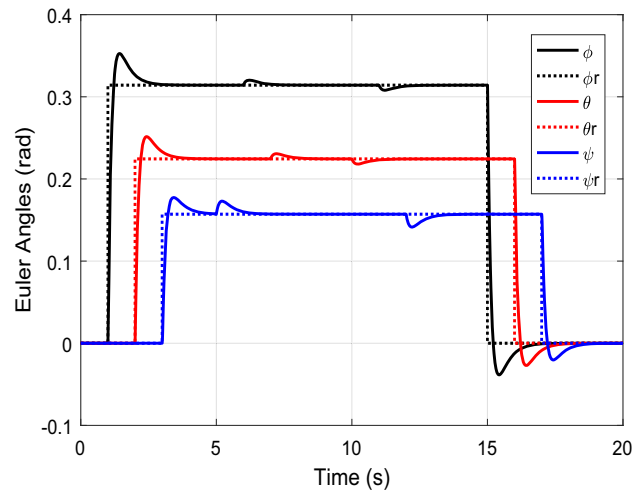

Fig. 6 Attitude system outputs by H_∞ control with attenuation level $\gamma = 1$ against the wind disturbances

Before the controller design stage, LPV modeling of the system has been done. For LPV modeling, the process described in Sect. (1–3) has been carried out and the initial Polytopic model (15) has been extracted. Then the PCA technique is programmed according to Sect. (2–3) and a reduced Polytopic model with 64 vertices is obtained. Then, by solving LMIs (28) and (33), the gains of the controller are obtained, and at the end, the system is simulated in the MATLAB/Simulink environment according to the block diagram in Fig. 4.

First, we consider three reference signals as pulses with different amplitudes values and different pulse durations, as the set points for Euler angles. Then implementing the proposed H_∞ control with disturbance attenuation γ, to the attitude system, the responses are shown in Fig. 5. It is observed that the responses have an appropriate damping coefficient, an acceptable percent of overshoot, and no steady state error.

Now implementing the wind effect to the proposed attitude controller. It is considered that wind disturbance torques about 1 *Nm* are arisen on the roll, pitch, and yaw axes at times t = 6 *s*, t = 7 *s*, and t = 5 *s* for 5, 3, and 7 s durations, respectively. Figure 6 shows the attitude angles. An excellent disturbance attenuation can be seen in this figure for the proposed H_∞ control with disturbance attenuation γ.

It is clear that the lower the attenuation level γ, the greater the disturbance attenuation. This can be well seen in Fig. 7, which plots the roll angle for different values of γ. Note that due to the clearer appearance and also similarity in behavior of all the three Euler angles, only the roll angle is drawn in this figure and subsequent figures.

As mentioned in the controller design section, we try to improve the controller performance by adding a pole assignment objective to the last controller. An LMI region $\mathcal{D}$ in the form of a 'conic sector' of the complex plane with inner angle

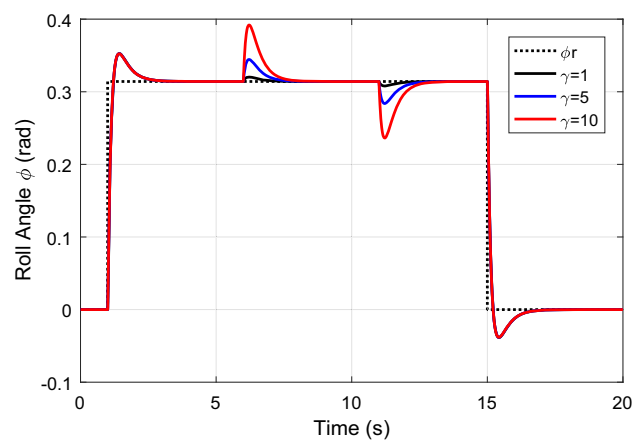

Fig. 7 Attitude roll angle by H_∞ control for different values for attenuation level γ against the wind disturbances

about 139°, it means damping ratio about 0.35, and apex at the $(-2\beta, 0)$, is considered for assigning the poles of the closed loop Polytopic LPV model. Figure 8 shows the roll angle response against the wind disturbances for different values for γ and β. As can be seen in this figure, for optimal values $\gamma^* = 0.8$ and $\beta^* = 12$, wind turbulence has not been able to affect the roll angle and deflect it. It should be noted that there is also a similar behavior for pitch and yaw angles, which has been avoided due to the increasing number of figures.

The control effort (angular torques) and the required speeds of the motors to produce this effort are shown in Figs. 9 and 10, respectively. It can be seen that the speeds of the motors belong to the predefined range specified in Table 1.

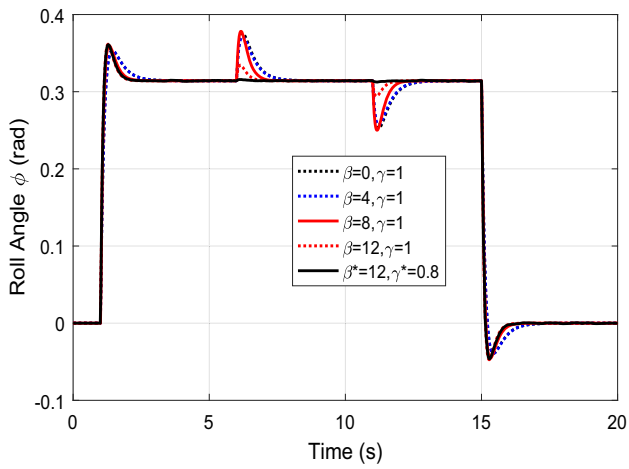

Fig. 8 Attitude roll angle by H_∞ control and pole placement for different values for γ and β against the wind disturbances

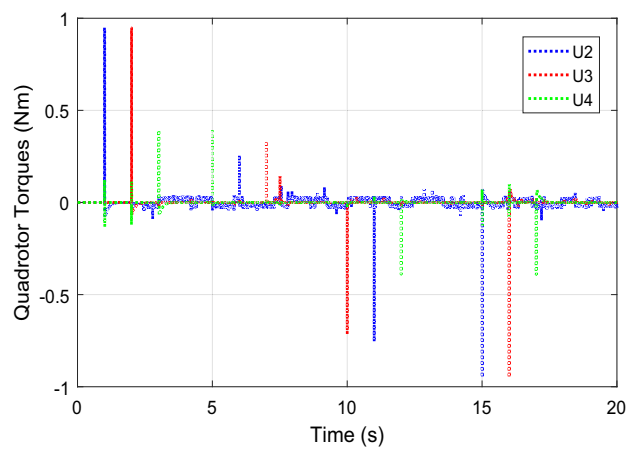

Fig. 9 Angular torques (control effort) applied to the quadrotor

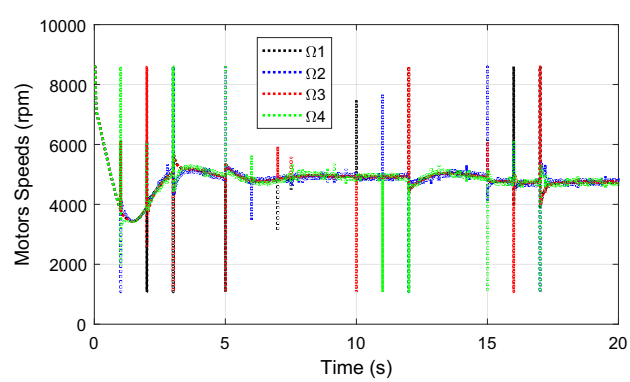

Fig. 10 Speeds of the motors

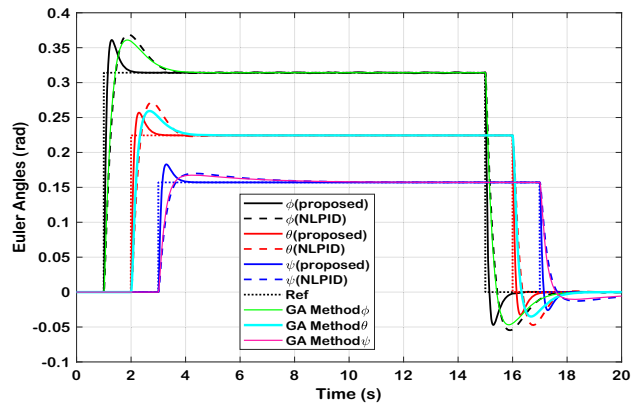

Fig. 11 Comparison of the proposed controller response with NLPID and GA response

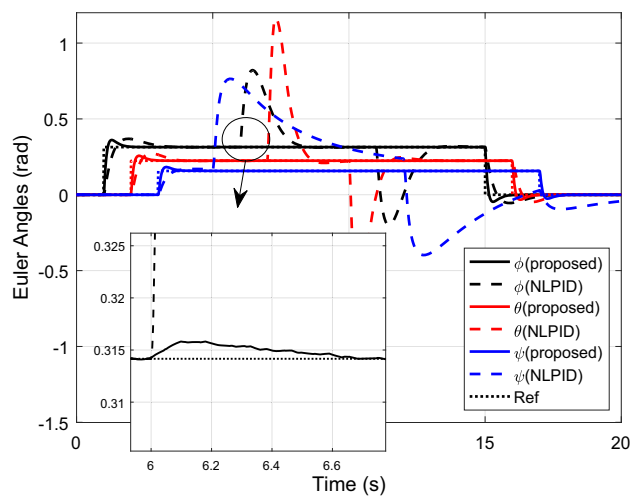

Fig. 12 Comparison of the proposed controller response with NLPID response in the presence of wind disturbances

The proposed method in [28] is considered to compare the capabilities of the proposed controller. In [28], nonlinear PID (NLPID) controllers are designed for roll, pitch, and yaw angles of a Quadrotor, where their parameters are tuned using Genetic Algorithm (GA) to minimize a multi-objective output performance index. To demonstrate how works parameters are tuned using (GA) is shown in Fig. 11. The Different types tuning of Algorithms selected parameters and Optimization methods are presented in [58–60].Fig. 11 compares our optimal case ($\gamma^* = 0.8$ and $\beta^* = 12$) response with NLPID and GA response. The very appropriate behavior of the proposed method toward NLPID can be easily seen in this figure. This performance is more highlighted in the presence of wind disturbances. For this reason, the effect of wind turbulence about 0.6 *Nm* on both controllers is shown in Fig. 12. It is observed that the wind disturbance was able to severely impair the NLPID response, while not having a significant effect on the proposed controller response.

5 Conclusion

In this paper, a method was proposed for Quadrotor attitude tracking control based on the structure of a typical PID controller, in the form of state feedback, using the H_∞ control theory in order to deal with external disturbances, which are mainly of the wind type and gyroscopic effects. By introducing suitable scheduling signals, a quasi-LPV modeling was introduced to cover the nonlinear effects of Quadrotor dynamics as well as its gyroscopic effects.

Gridding the scheduling signals in their variation ranges, an initial high-dimensional Polytopic model is obtained. Due to the high number of vertices of the initial Polytopic model, the parameter set mapping is implemented based on the PCA algorithm to achieve a tighter region in the parameter space. The proposed control is derived from a typical PID controller in the form of a state feedback control, except that it has a centralized structure and not a decentralized one. Therefore, it is definitely superior to the traditional PID control that is applied to each axis separately; while, it is easy to consider extra control objectives, which here has been the disturbance rejection and pole placement. The proposed control has a simple structure and low consuming computation time because it is in fact same PID with interactive in all axes. Although the proposed control design may seem complicated and time-consuming, because of the drawback related to the increase in the LMIs and the increase in the computation volume, but it should be said that in the proposed controller all the LMIs are solved off-line and its computing time is not a problem in the real time implementation. However, there has been extensive research on the computational complexity of LMIs so far [61, 62]. As future works, the following can be mentioned. The proposed control strategy can be expanded on Quadrotor maneuver control, where in addition to the attitude control, the position control and also the tracking of a desired path is the main problem. Another suggestion is to improve the proposed control from a fixed control and upgrade it to a variable structure dependent on scheduling variables. In this way, the conservatism of the proposed control will be significantly reduced.

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Author contributions The author contributions can be summarized to: Considering the gyroscopic effects and other nonlinearities of the system in the Polytopic modeling. Using PCA algorithm to reduce the size of Polytopic model data. Providing the proposed control under the PID structure for easier practical implementation.

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Data availability Data availability is not possible.

Code Availability Code Availability is not applicable.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

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